

Chapter 14 — Literature

Finding, screening and citing without fabricating

The highest-demand capability in this room, and the one with the most ways to go quietly wrong.

Start from the failure gallery

You have already seen a fabricated citation from your own field, and what it took to catch it.

- It was fluent. It had plausible authors, a plausible year, and a plausible venue.
- It was caught by resolving the identifier, not by reading the sentence.
- Chapter 5 explained why: the middle band of model knowledge, where it believes it knows.

Everything in this chapter is downstream of that one demonstration.

What survives PDF extraction, and what does not

- Two-column layouts interleave. Text from the left column and the right can arrive as one sentence that neither column contains.
- Equations degrade badly. Sub- and superscripts flatten, symbols are substituted, and the result is often syntactically plausible and numerically wrong.
- Tables lose their structure. A number can migrate a column.

Check the extraction before you reason about the content. A wrong number extracted cleanly looks exactly like a right one.

This is Chapter 4's discipline applied to a file format: verify the intermediate step before interpreting the result.

Reference managers and the document pipeline

- Keep one source of truth for bibliography data, and export from it. Retyping a reference is how a wrong year enters a paper.
- Choose an export format that survives your document pipeline, and test it on one reference before you have three hundred.
- The identifier — DOI, arXiv ID — is the field that matters. Author and year are convenience; the identifier is the thing you can check.

Screening against stated criteria

- Write the inclusion and exclusion criteria **before** you screen. Same rule as the eval in Chapter 4, and for the same reason.
- Screening is a classification task with a defined answer, which makes it one of the few literature tasks that can be checked.
- **Papers get silently dropped.** A screening pass returns what it returns; it does not report what it failed to consider.

So screen a known subset first — twenty papers where you already know the answer — and measure the false-negative rate before trusting it on the ones you have not read. That is a five-case eval wearing different clothes.

Summarising without laundering

- A summary that drops the authors' stated scope converts a bounded finding into a general claim. That is laundering, and it is usually accidental.
- The scope sentence is the one most often lost: which models, which dataset, which conditions.
- **Keep the hedge.** If the authors wrote "suggests", you do not get to write "shows".

You have seen this failure twice in this course already — a prompting result that held on one model family being quoted as general, and a hardware result true on a benchmark being quoted as deployed.

The habit that closes the thread

Every citation is verified against a primary identifier. No exceptions.

- Resolve the DOI or the arXiv ID. Open the record. Confirm the title, the authors and the year.
- A citation you have not resolved is a candidate, not a reference.
- This costs about thirty seconds per source and it is the entire defence.

Thread — truthfulness, closed. Introduced in Chapter 2 as a rated rubric dimension, explained in Chapter 5 as a structural property rather than a bug, and enforced here as a work habit you can actually perform.

Exercise

1. Take a paper you know well. Extract it and check what survived — equations first.
2. Ask for a summary. Find the scope sentence in the original and check whether it survived.
3. Ask for five references on a narrow question in your field. **Resolve every identifier**. Record how many were real, how many were real but said something else, and how many did not exist.

Step 3 is the one you will remember. Write the numbers down — they are your own failure gallery, and they are more persuasive to you than anyone else's.